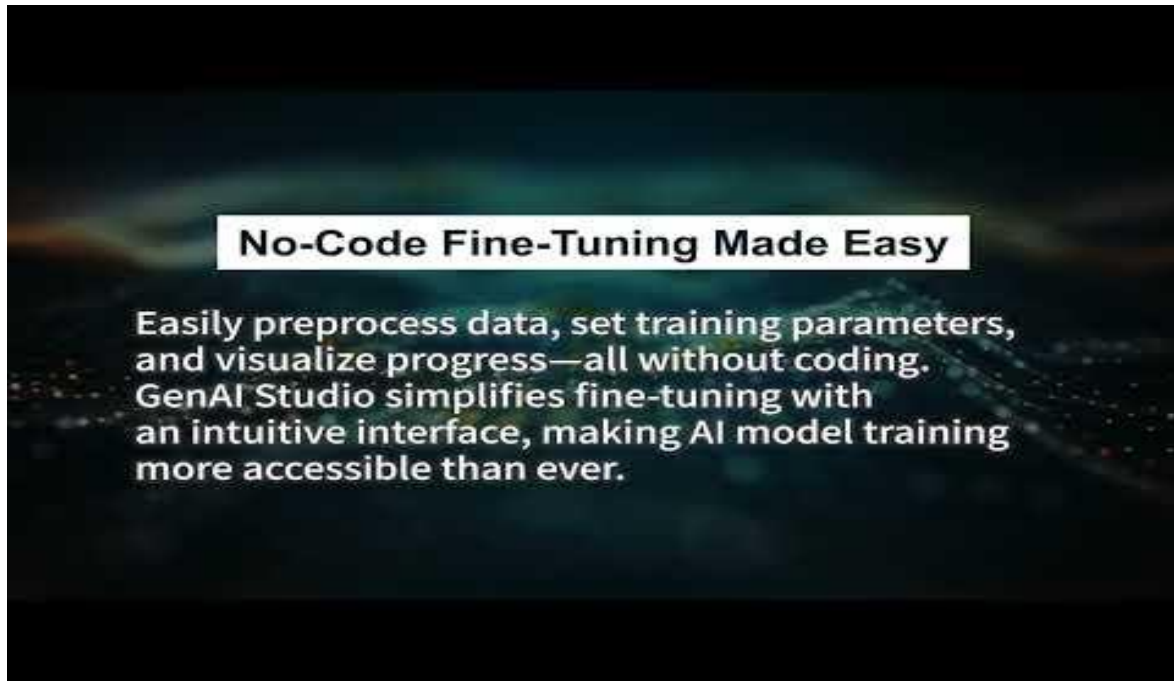


How to finetune LLM-model Gen-ai

Learn how to setup Gen-AI studio and finetune LLM-models on your computer.

First take a look at this video: <https://youtu.be/GVMexUe7DDA>

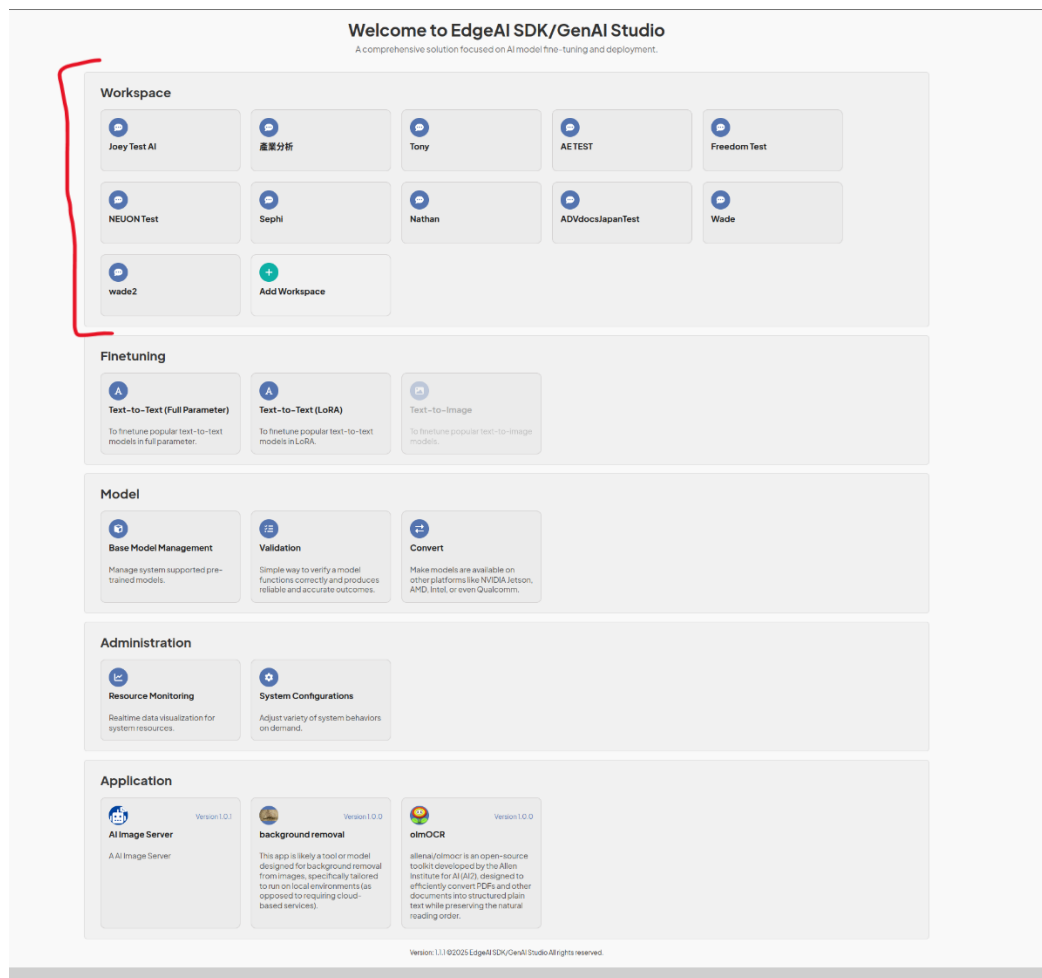


1. Get Gen-ai environment

- Go to <http://218.32.15.226:3001/login>
 - ID : ajp-fae
 - PW : ajp@0409

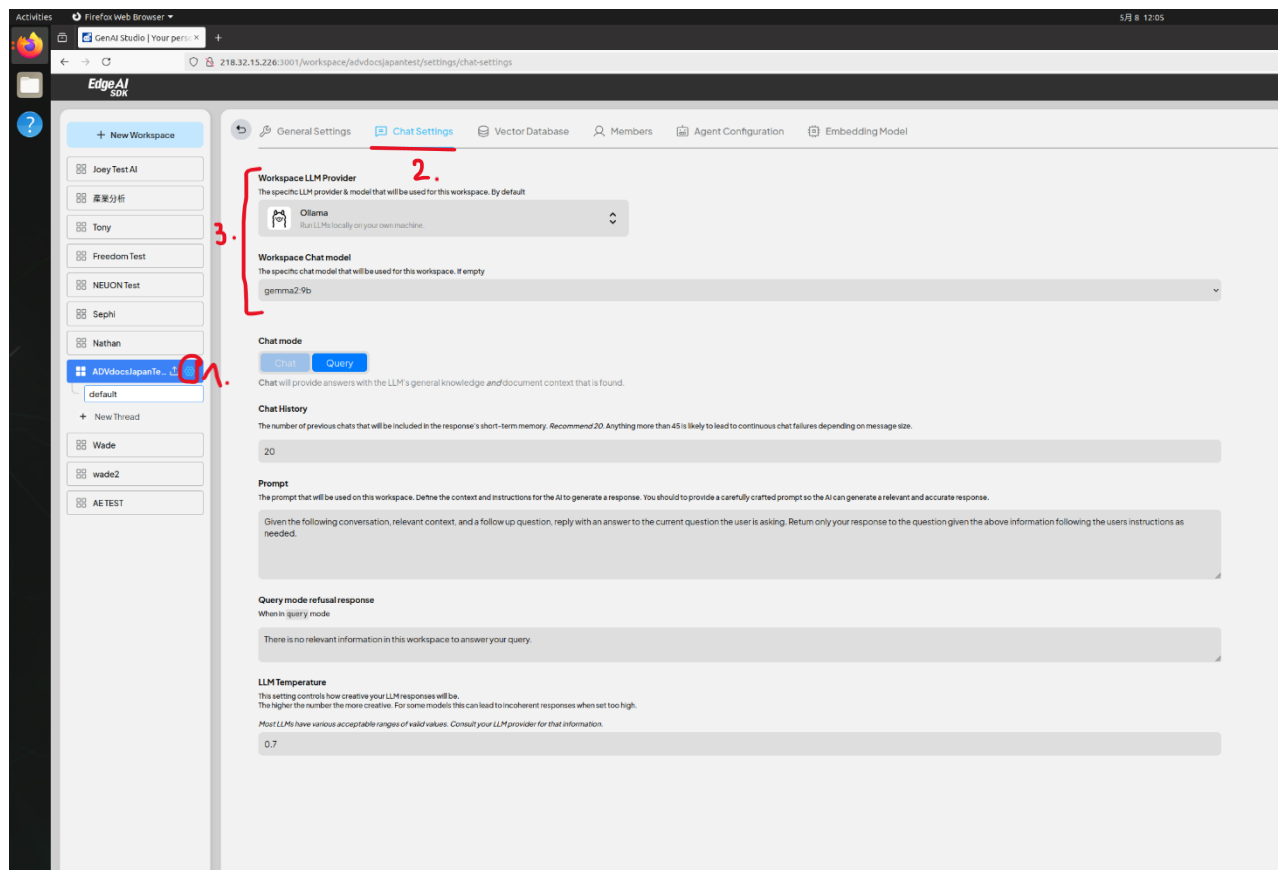
2. Test workspace

- Go to a workspace or create one and test the LLM-model.



You can create a new workspace or use an existing one. This step is just for checking if the environment is functioning.

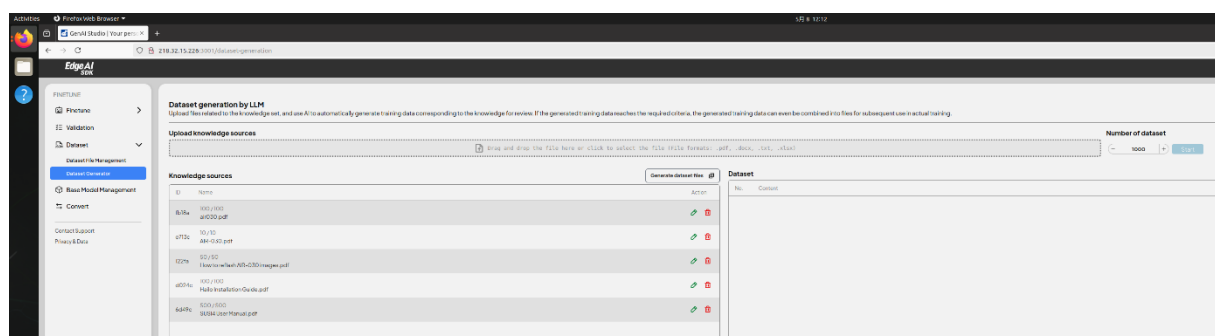
- Press cartwheel(settings) icon of the workspace and go to Chat settings
 - Here you can choose your preferred LLM-provider and chat model



Currently, only Ollama LLM provider works due to a token issue. Feel free to check out if the others work. Otherwise use Ollama.

3. Generate Dataset

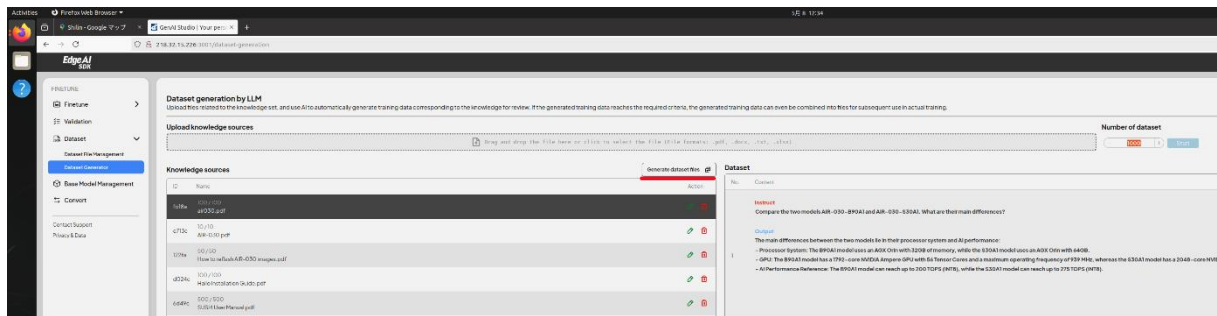
- From home go to finetuning menu and select dataset generator



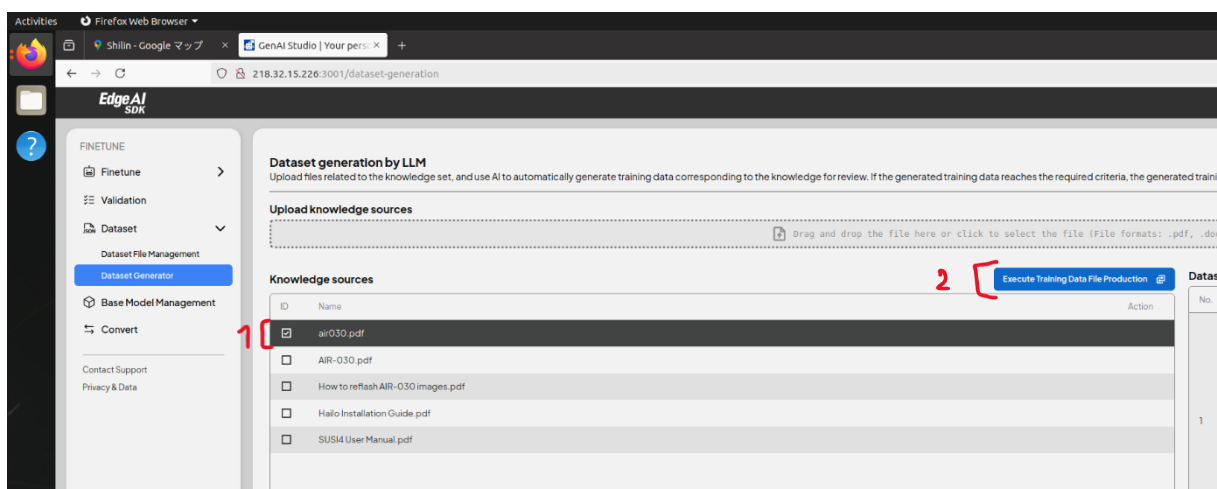
This step is to create a JSON-file which will consist of questions the environment will make for the model to train on. This is our training data.

- Upload document that needs to be trained(Upload knowledge sources)
- And select number of questions(number of dataset)

- After finished press generate dataset files and select the right set



The higher the number of dataset, the more accurate or fine-tuned your model will be.



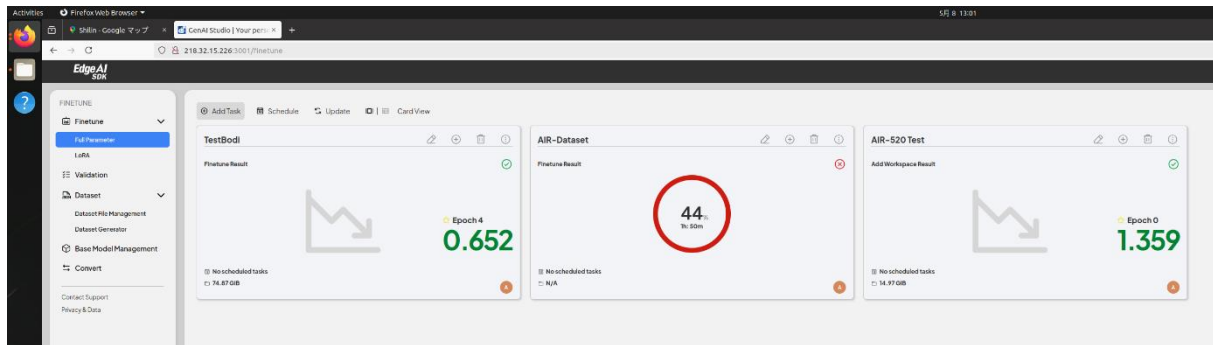
Choose the documents you want the model to be trained upon. You can add multiple documents but note that each document can be max 10 MB.

- Enter Dataset name and confirm

For good-practice, please try to use the same naming-convention as the others.

4. Finetune model

- Make a new task and select model and created dataset



A task consists of fine-tuning a model.

- Edit parameters and start training

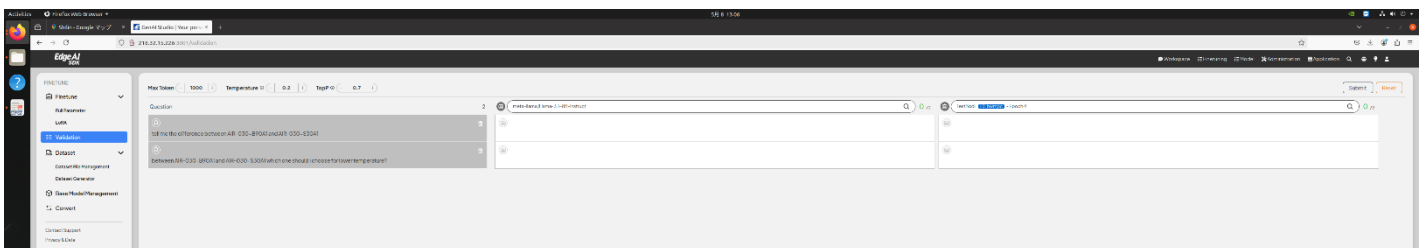


5. Validate model

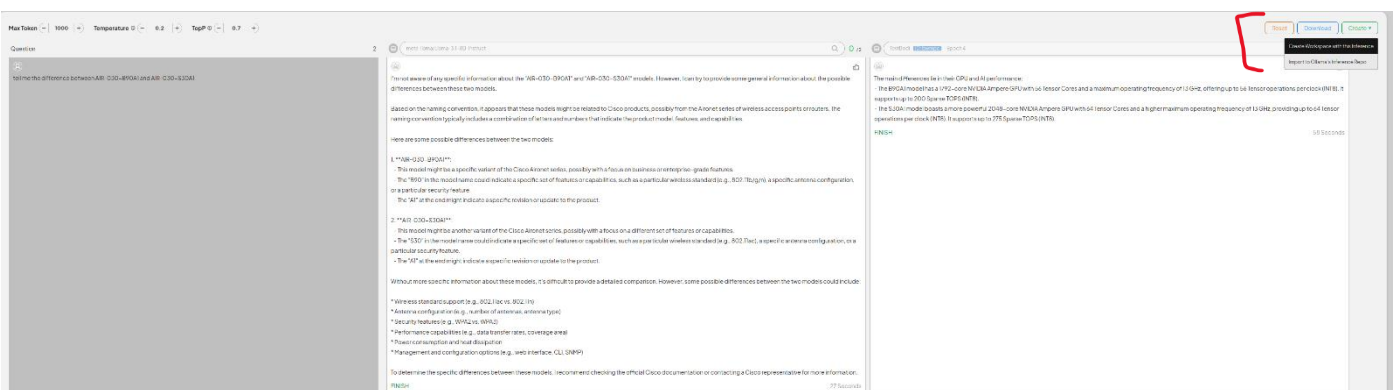
- Go to validation tab and select the normal model and the fine tuned model

The image shows two side-by-side "Select Model & Epoch" dialog boxes. The left dialog has "Base Model" selected, with a dropdown menu showing "meta-llama/llama-3.1-8B-Instruct". Below it, "Finetune" is selected, with "Category" set to "Full Parameter", "Task" set to "AIR-520 Test", "Finetune" set to "Llama-3.1-8B-Instruct / air-ds_1k_4prompts_new.json", and "Epoch" set to "0". The right dialog has "Finetune" selected, with "Category" set to "Full Parameter", "Task" set to "TestBodi", "Finetune" set to "Llama-3.1-8B-Instruct / air030.json", and "Epoch" set to "4". Both dialogs have "Cancel" and "Confirm" buttons at the bottom.

- Ask a couple of questions and see if the fine-tuning



- Create workspace with new fine-tuned model



Your finetuned model should work now.